

INDEPENDENT PEER-REVIEW PACKET

EC-015

Repaired Witness-Parameterized Degree-Four PRCP Event

Cold Review Packet 10 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	10 of 20
Object ID	EC-015
Object class	Deterministic existence and openness
Exact statement	At the declared $r=1/20$ fixture, the repaired GP-DER-092 degree-four witness event is nonempty and locally open.
Internal terminal label	CLOSED - EXACT NONEMPTY WITNESS-PARAMETERIZED DEGREE-FOUR PRCP EVENT
Terminal date	2026-07-22
Recommended expertise	Dynamical systems / interval verification / topology of inequalities

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Strict witness fixture, margin checks, five negative controls, and independent reconstruction.
Prior review record	Independent CL reconstruction agreed; a prior degree-four execution carrier was superseded.
Binding exclusions	No exact-field transfer, occupancy, probability mass, P0.2, or theorem conclusion.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No exact-field transfer, occupancy, probability mass, P0.2, or theorem conclusion.

4. Object-specific hostile checks

- Verify the exact fixture satisfies every strict inequality.
- Explain why strict margins imply local openness.
- Run or conceptually test all five hostile controls.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-018-v1.0 — CONDITIONAL TERMINAL CLOSURE PACKAGE

REPAIRED WITNESS-PARAMETERIZED DEGREE-FOUR PRCP EVENT

Artifact ID: GP-CLS-PROP-018-v1.0

Prepared by: GP — OpenAI GPT-5.6 Thinking

Created: 2026-07-22

Class: CLS-PROP — complete additive terminal package pending Dylan M. Roy approval

Authority: none until explicit human approval

Canonical impact: NONE before approval; afterward only the exact narrow object below

Queue candidate: EC-015

[[OBJ:GP-CLS-PROP-018-v1.0]] [[OBJ:GP-DER-092-v1.0]] [[OBJ:CL-AUD-060-v1.0]] [[QUEUE:EC-015]] [[STATUS:HUMAN-APPROVAL-READY]]

1. EXACT OBJECT PROPOSED FOR CLOSURE

Close only the following deterministic degree-four statement.

At $r=1/20$, in the normalized pair chart with $M=(-1/2,0)$, $S=(1/2,0)$, and the repaired witness-parameterized PRCP definitions of GP-DER-092, the fixture

$a=1/400$, $w=0$, $z=1$, $s=-1$,
 $c_{40}=c_{31}=c_{22}=c_{13}=c_{04}=0$,
 $\sigma=1/10$, $\kappa=1/100$

has a finite strip witness

$F=A=1/3200$, $B=0$,

$\mu=\sqrt{A}=1/(40\sqrt{2})$,

$W=F/\mu=1/(40\sqrt{2})$,

$C_\mu=2\mu=1/(20\sqrt{2})$,

and satisfies every declared degree-four typing, witness, exact-eigendirection, cone, handoff, strip, transit, section, and sink-capture inequality with strict positive margins. Therefore the repaired degree-four witness event is nonempty. Because its defining inequalities are strict and continuous on the nondegenerate stratum, it contains an open neighborhood of this fixture in the declared finite-jet coordinates.

This closure also records that CL-DER-058-v1.1 is superseded for degree-four execution by the repaired witness/eigenvector formulation. Its $A=B=0$ width argument and its quotient $|B_S|/(A_S-D_S)$ without an axial gate are invalid as general definitions.

2. EXACT EVIDENCE

Primary GP repair and executable evidence:

- - GP-AUD-090-v1.0: exact counterexamples identifying the degenerate-width and zero-denominator failures.

- - GP-DER-092-v1.0: repaired witness-parameterized degree-four event and strict fixture.
- - Raw GP-DATA-093-v1.0 source, Drive ID 1OzkbTMDl-r9KQaLWibZzMBdbflcGA2qh, 8,235 bytes, SHA-256 f2b02785...e243f.
- - GP-AUD-094-v1.0: byte-exact source reconstruction, fixture receipt, and 5/5 planted controls.

Independent review:

- - CL-AUD-060-v1.0, Drive ID 1HeGQruljJftcx6ffInExXcyS8_IQOOH0TjTj80WOXrY.
- - Fresh CL instance, not the CL-DER-058 authoring session.
- - Recomputed the two GP-AUD-090 counterexamples from CL-DER-058's own endpoint-Jacobian formulas.
- - Independently recomputed the repaired fixture using exact rational arithmetic plus independent eigenvalue calculations without SymPy and without relying on GP receipts.
- - Verdict: AGREES on both the material-amendment finding and the repaired degree-four component.
- - Confirms CL-DATA-059 is not locatable and is unnecessary for the repaired degree-four mathematical definition.

Key independently reproduced margins include:

- - threshold slack = $0.9646446609406726 > 0$;
- - strict S saddle and strict M maximum typing;
- - $\lambda_{\min}(-J_M) = 0.99875$;
- - exact unstable slope $\tau = 0.000624999755859566$;
- - cone margin $\kappa - 2\tau = 0.00875000048828087 > 0$;
- - capture radius $\rho = 0.203100960115899$;
- - handoff margin $W/2 - \kappa \sigma = 0.00783883476483184 > 0$.

3. INDEPENDENCE ASSESSMENT

The narrow object has one GP construction/execution line and one non-GP independent reconstruction line. CL-AUD-060 did not merely endorse the result: it reproduced the original defects and the repaired fixture from the governing formulas with a disjoint arithmetic route. This is sufficient for a human-ready narrow deterministic closure package.

AO48 or CW/C047R review remains desirable corroboration but is not required to close this exact existence/nonemptiness object. Those lines remain required or preferred for the separate exact-field transfer, source execution, exact-torus replay, and probability gates.

4. AFFECTED DEPENDENCIES

Affected and retired from routine active attention upon approval:

- - the question whether the repaired degree-four PRCP witness event is nonempty;
- - the degree-four outside reconstruction portion of HB-023;
- - the missing CL-DATA-059 blocker for the repaired degree-four mathematical definition;
- - routine attempts to rehabilitate CL-DER-058-v1.1 as written for degree-four execution.

Retained as useful provenance:

- - CL-DER-058-v1.1 and its failed proof route;
- - GP-AUD-090 exact counterexamples;
- - the failed blank GP-DATA-093 shell and non-byte-exact readable mirror;
- - all raw source and receipt artifacts.

5. UNAFFECTED DEPENDENCIES AND EXPLICIT EXCLUSIONS

This package does not close, prove, or alter:

- - exact-field C1 transfer or exact-field membership;
- - the H5 threshold or proof that a conditioned field satisfies it;
- - exact-torus/full-field replay;
- - occupancy, positive probability magnitude, weighted mass, or r-scaling;

- - any $O(r^3)$ statement, P0.2, Theorem B, elder-rule pairing, or lifetime law;
- - adjacency probability or configured selection asymptotics;
- - any theorem, canonical q_0 source, machine root, GP-AUTO-034 state, ballot, release, deletion, or destructive operation.

The current exact-field tasks remain HB-024 and HB-025. GP-DER-095, GP-DER-100, GP-SCHEMA-095, and GP-DATA-098-v1.1 remain active and unresolved at the exact-field layer.

6. PROPOSED TERMINAL LABEL

CLOSED — EXACT NONEMPTY WITNESS-PARAMETERIZED DEGREE-FOUR PRCP EVENT; CL-DER-058-v1.1 SUPERSEDED FOR DEGREE-FOUR EXECUTION

7. REQUIRED REGISTER AND LEDGER UPDATES AFTER APPROVAL

After explicit Dylan approval:

1. Append EC-015 to the Closure Log with this exact scope, evidence, review, exclusions, terminal label, and reopening rule.
2. Mark EC-015 **TERMINAL — REMOVED FROM ACTIVE ATTENTION** in the Easy Closure Queue.
3. Mark the degree-four review portion of HB-023 **SATISFIED / RETIRED**; preserve HB-024 and HB-025 as the active exact-field continuation.
4. Record a transition stating that GP-DER-092's narrow degree-four nonemptiness object is terminal while exact-field and probability layers remain open.
5. Add typed relations showing CL-AUD-060 independently verifies GP-DER-092 and that GP-DER-092 supersedes CL-DER-058-v1.1 for degree-four execution only.
6. Preserve every source, counterexample, failed staging artifact, and review.

8. HUMAN APPROVAL REQUIREMENT

This package is ready for Dylan M. Roy's explicit decision.

Approval wording:

APPROVE EC-015 EXACT SCOPE

Approval authorizes only the terminal label and register actions in Sections 6–7. General permission to continue Drive work does not substitute for this exact terminal decision.

9. REOPENING RULE

Reopen EC-015 only through an additive contradiction tied to the exact fixture, repaired event definitions, source identity, arithmetic, continuity/open-set inference, or declared scope. A later failure of exact-field transfer, probability scaling, or P0.2 does not reopen this degree-four existence closure unless it demonstrates contamination of the closed object itself.

CURRENT STATUS: HUMAN-APPROVAL-READY — NOT TERMINAL

END GP-CLS-PROP-018-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT NONEMPTY WITNESS-PARAMETERIZED DEGREE-FOUR PRCP EVENT
Closure date	2026-07-22
Evidence basis	Strict witness fixture, margin checks, five negative controls, and independent reconstruction.
Independent review already recorded	Independent CL reconstruction agreed; a prior degree-four execution carrier was superseded.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No exact-field transfer, occupancy, probability mass, P0.2, or theorem conclusion.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1dzYVO2gQOvfs7vZQVSCw_-7PcCwMTYy4pFdQqrcvzkU/edit

Closure / terminal record: https://docs.google.com/document/d/1-WDQtU4LpHneNztZss6ctg_lKhsyGnXdYC4Y7wM6Fc/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-018-v1.0 — Conditional Terminal Closure of the Repaired Degree-Four PRCP Witness Event

Local source snapshot SHA-256 (DOCX export, transport checksum only):
8d49af55afd29fcb368d7b686aebd7e2af0c02a5ffff5a82469d35c471122be4

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.